GOOD NOTATION

THE DATA AND AI SCHOOL

Agentic Ai Profe ssional

From Zero to Production

A structured journey from Python fundamentals to autonomous multi-agent AI systems you can actually ship.

12	4	12
	PHASES	

PHASE 1

Foundation

MasterthebuildingblockseveryAI system is built on.

PHASE 1 — FOUNDATION

01 · Python + APIs

02 · SQL + Data Layer

PHASE 2 — INTELLIGENCE

03 · LLM Fundamentals

04 · AI Agent Design

05 · Tool Integration

06 · Memory Systems

PHASE 3 — SYSTEMS

07 · Multi-AgentSystems

08 · RAG Systems

09 · Automation

PHASE 4 — PRODUCTION

10 · Deployment

11 · Monitoring &Security

12 · Capstone Project

MODULE 01

Python + APIs

BACKEND

Master Python and learn to communicate with the world via APIs.

// LESSON TOPICS

- Python environment setup, virtual envs & project structure
- Functions, classes, decorators & object-oriented programming
- REST API concepts — GET, POST, PUT, DELETE & HTTP status codes
- HTTP clients: requests library & async httpx for high-performance calls
- Authentication: API keys, OAuth2, Bearer tokens & session handling
- Error handling, exponential retries & rate limiting strategies
- Async programming with asyncio & aiohttp for concurrent requests

//LEARNINGOUTCOME

By the end you canwrite a robust Python backend that calls any external API, handles errors gracefully, and processes JSON data reliably — the foundation every AI agent is built on.

PROJECT	**Build a multi-source data aggregator that pulls from 3 live APIs and presents unified results**

// TECH STACK

Python 3.11+ · requests · httpx · FastAPI · Pydantic · asyncio

MODULE 02

SQL + Data Layer

DATA

Design databases that store agent memory and conversation history.

// LESSON TOPICS

- Relational database design principles & normalization (1NF–3NF)
- SQL mastery: SELECT, JOIN, GROUP BY, window functions & subqueries
- SQLite for local development, PostgreSQL for production workloads
- SQLAlchemy ORM — models, sessions, relationships & Alembic migrations
- Storing conversation logs, agent state & user sessions persistently
- pgvector extension — vector storage and similarity search for embeddings
- Query optimization: indexes, EXPLAIN ANALYZE & connection pooling

// LEARNING OUTCOME
You will design andquery databases that persist agent memory, logs, and user data — enabling your AI systems to remember context across sessions and scale to millions of records.

PROJECT	**Design a full conversation memory database with CRUD, search, and analytics queries**

// TECH STACK
SQLite · PostgreSQL · SQLAlchemy · Alembic · pgvector · psycopg2

PHASE 2

Intelligence

UnderstandLLMsdeeplyandbuild agents that reason and act.

MODULE 03

LLM Fundamentals

AI CORE

Understand how language models think and how to steer their behavior precisely.

// LESSON TOPICS

- Transformerarchitecture, attention mechanisms & how LLMs generate tokens
- Tokens, context windows, temperature, top-p & sampling strategies
- Prompt engineering: zero-shot, few-shot, chain-of-thought & structured output
- System prompts, personas, role design & instruction hierarchy
- OpenAI SDK deep-dive: completions, function calling & streaming
- Anthropic Claude SDK: messages API, tool use & vision capabilities
- Model selection trade-offs: GPT-4o vs Claude vs Gemini vs open-source
-
-

// LEARNING OUTCOME
You will understandLLMs from the inside out — knowing exactly why they behave as they do, and crafting prompts that reliably produce the precise outputs your applications require.

PROJECT

Build a prompt optimizer that uses an LLM to self-improve its own responses via reflection loops

// TECH STACK

OpenAI SDK · Anthropic SDK · tiktoken · LangChain · PromptFlow · Instructor

MODULE 04

AI Agent Design

AGENTS

Build agents that reason, plan, and act autonomously without constant human input.

// LESSON TOPICS

- Agent architectures: ReAct, Plan-and-Execute & Reflection patterns
- The think → act → observe → reflect loop in detail
- Tool calling & function calling APIs — design and implementation
- Goal decomposition, task planning & hierarchical sub-goal management
- Self-reflection, error correction loops & graceful failure recovery
- LangChain Agents deep-dive & LangGraph for stateful agent workflows
- Guardrails: stopping conditions, safety checks & output validation

// LEARNING OUTCOME

You will build agents that autonomously break down goals, select appropriate tools, observe results, and adapt their strategy — the core skill of modern AI engineering.

PROJECT

Build a research agent that plans a research strategy, searches the web, and writes a structured report

// TECH STACK

LangChain · LangGraph · OpenAI Functions · Pydantic AI · Instructor · Guardrails AI

MODULE 05

Tool Integration

INTEGRATION

Give your agents hands — let them take real actions in the world.

// LESSON TOPICS

- Designing tool schemas that agents can reliably understand and invoke
- Web search tools: Tavily, Serper & Bing Search API integration
- Code execution sandboxes: E2B & Modal for safe code running
- File and document manipulation: read, write, parse PDFs & spreadsheets
- Browser automation with Playwright for web scraping & form filling
- Email, calendar & Slack integrations via official APIs
- Building custom tools from any third-party API in minutes

//LEARNINGOUTCOME

Your agents will move beyond text — searching the web, executing code, reading documents, and interacting with real-world services on your behalf around the clock.

PROJECT

Build a personal assistant agent equipped with 6+ real-world tools that can complete multi-step tasks

// TECH STACK

Tavily · E2B · Playwright · Gmail API · Slack SDK · Composio · PyMuPDF

MODULE 06

Memory Systems

MEMORY

Give agents long-term memory that persists across sessions and personalizes to each user.

// LESSON TOPICS

- Types of agent memory: short-term, long-term, episodic & semantic memory
- In-context memory management and conversation compression strategies
- Embeddings — what they are, how to generate them & when to use them
- Vector databases in depth: Pinecone, Weaviate & Chroma setup and querying
- Semantic similarity search, cosine distance & relevance scoring
- Memory summarization, forgetting curves & knowledge compression
- User-specific personalized memory stores for multi-tenant applications

// LEARNING OUTCOME

You will implementmemory systems so agents remember users, learn from past conversations, andretrieve precisely relevant knowledge — making them genuinely more intelligent over time.

PROJECT	**Build an agent with persistent cross-session user memory that personalizes responses based on history**

// TECH STACK

OpenAI Embeddings · Pinecone · ChromaDB · Weaviate · Mem0 · LangMem

PHASE 3

Systems

Orchestratecomplexmulti-agent workflows and knowledge pipelines.

MODULE 07

Multi-Agent Systems

ARCHITECTURE

Orchestrate teams of specialized agents working together on complex problems.

// LESSON TOPICS

- Multi-agentpatterns: supervisor, swarm, pipeline & hierarchical architectures
- Agent communication protocols & message passing between agents
- LangGraph for building stateful, cyclical multi-agent graphs
- Parallel execution strategies & intelligent task delegation
- Conflict resolution, consensus mechanisms & voting between agents
- CrewAI framework for role-based multi-agent teams
- Shared memory stores, coordination strategies & avoiding deadlocks

// LEARNING OUTCOME

You will design systems where specialized agents collaborate in parallel — a researcher, writer, critic, and editor working in concert to produce outputs no single agent could achieve alone.

PROJECT

Build a content production pipeline with 4 specialized agents collaborating to research, draft, edit, and publish

// TECH STACK

LangGraph · CrewAI · AutoGen · Ray · Celery · Redis · RabbitMQ

MODULE 08

RAG Systems

RAG

Give agents access to massive knowledge bases with retrieval-augmented generation.

// LESSON TOPICS

- RAG architecture explained: index, retrieve, augment & generate pipeline
- Document loaders, text chunking strategies & optimal chunk sizing
- Embedding models comparison & dense vs sparse retrieval methods
- Hybrid search: combining BM25 keyword search with semantic similarity
- Re-ranking with Cohere Rerank & cross-encoder models for precision
- Advanced RAG techniques: HyDE, RAG Fusion, FLARE & self-RAG
- Evaluating RAG quality: faithfulness, context recall & answer relevance with RAGAS

// LEARNING OUTCOME

You will build RAGpipelines that let agents answer questions accurately over thousands of documents — powering enterprise knowledge bases, intelligent document Q&A, and research assistants.

PROJECT

Build a full RAG system over a 500-page technical document corpus with hybrid search and re-ranking

// TECH STACK

LlamaIndex · LangChain · Pinecone · Cohere · RAGAS · Unstructured · sentence-transformers

---

MODULE 09

Automation

AUTOMATION

Automate complex real-world business processes end-to-end with AI at the core.

// LESSON TOPICS

- Workflow design principles: triggers, conditions, actions & branching logic
- Webhook ingestion, event-driven architectures & message queues
- n8n & Make (Integromat) for visual no-code/low-code automation
- Scheduled jobs with cron syntax, APScheduler & Celery Beat
- Error recovery patterns, dead letter queues & idempotent processing
- Human-in-the-loop approval workflows & conditional escalation
- Building end-to-end AI-powered data transformation pipelines

//LEARNINGOUTCOME

You will automatecomplete business workflows that run 24/7, handle failures gracefully, scale automaticallywith load, and know exactly when to ask a human for help.

PROJECT

Build a complete lead qualification and personalized outreach automation pipeline with AI scoring

// TECH STACK

n8n · Apache Airflow · APScheduler · Temporal · FastAPI Webhooks · Redis Streams

PHASE 4

Production

Ship your AI systems to the cloud reliably, securely, and at scale.

MODULE 10

Deployment

DEVOPS

Package and ship your AI systems to production cloud infrastructure reliably.

// LESSON TOPICS

- Containerizing AI agents with Docker — Dockerfiles, layers & best practices
- Deploying to AWS ECS, Google Cloud Run & Railway for managed containers
- CI/CD pipelines with GitHub Actions — automated test, build & deploy
- Environment management, secrets handling & 12-factor app principles
- Load balancing strategies, auto-scaling & horizontal scaling patterns
- Serverless functions with AWS Lambda for event-driven agent endpoints
- Blue/green deployments, canary releases & instant rollback strategies

//LEARNINGOUTCOME

You will package and deploy your AI systems to production cloud infrastructure with automated testing, one-command deploys, zero-downtime updates, and auto-scaling under load.

PROJECT	**Deploy a full agent system with CI/CD pipeline to AWS with auto-scaling and health monitoring**

// TECH STACK

Docker · AWS ECS · Google Cloud Run · Railway · GitHub Actions · Terraform · Nginx

MODULE 11

Monitoring & Security

ENTERPRISE

Make your AI systems observable, secure, and trustworthy for enterprise use.

// LESSON TOPICS

- LLM observability platforms: LangSmith & Langfuse for full run tracing
- Distributed tracing of agent runs end-to-end across services
- Cost tracking dashboards, token usage monitoring & budget alerts
- Prompt injection attacks, jailbreak techniques & defense strategies
- Input validation, output sanitization & content policy enforcement
- Rate limiting, API authentication, JWT tokens & role-based access
- Alerting on failures, latency spikes, cost anomalies & error rates

// LEARNING OUTCOME

You will add the observability and security layers that enterprise production demands — seeing exactly what your agents do, catching failures within seconds, and defending against adversarial inputs.

PROJECT	**Add full observability stack and security hardening to your production agent with custom dashboards**

// TECH STACK

LangSmith · Langfuse · Prometheus · Grafana · Sentry · Guardrails AI · OpenTelemetry

MODULE 12 · CAPSTONE

Full AI System

Combineeverythingintooneproduction-grade autonomous AI system.

Combine everything from all 12 modules into one complete, production-grade autonomous AI system. You will architect, build, deploy, monitor, and present a full end-to-end solution that demonstrates mastery across every skill in this course.

// DELIVERABLES

ARCHITECTURE Systemdesign doc with full agent flow diagram	BACKEND PythonAPIs + PostgreSQL + vector database
AGENT CORE Multi-agentsystem with RAG + persistent memory	INTEGRATIONS 3+real-worldtool & API connections
DEPLOYMENT Liveoncloud with CI/CD pipeline	OBSERVABILITY Fullmonitoring, tracing & alerting dashboards
DEMO Livepresentation + technical walkthrough	PORTFOLIO Open-source GitHub repo + polished README

// EVALUATION CRITERIA

- Technical Completeness — does the system cover all required components?
- Code Quality — is the codebase clean, modular, and well-documented?
- Agent Intelligence — does the system reason, plan, and adapt effectively?
- Production Readiness — is it deployed, monitored, and secure?
- Presentation — can you clearly explain architecture decisions and trade-offs?

GOOD NOTATION
THE DATA AND AI SCHOOL

Address: Tech Zone IV, Amrapali Riverview Greater Noida West ,
Uttar Pradesh 201318
Contact Us : +91 92891 70269